SARDAR UMAIR NAWAZ KHAN

AI Systems Engineer | Machine Learning & Agentic Automation

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PROFESSIONAL SUMMARY

Computer Science undergraduate at FAST-NUCES specializing in Agentic AI systems, Machine Learning architectures, and high-performance backend pipelines. Experienced in training predictive models and neural pipelines with PyTorch and Scikit-learn, engineering autonomous multi-tool LLM workflows with LangChain and n8n, designing asynchronous RESTful APIs using FastAPI, Node.js, and PHP, and implementing normalized relational databases in PostgreSQL and MySQL. Proven track record of building production-oriented full-stack software and resilient automation systems.

TECHNICAL SKILLS

Programming Languages: Python, Modern C++ (C++17), TypeScript, JavaScript (ES6+), PHP, SQL, Bash

AI, Generative AI & ML: PyTorch, Scikit-learn, Generative AI, LLM Integrations, Fine-Tuning, Autonomous AI Agents, LangChain, n8n Orchestration, RAG Workflows, PCA, Hierarchical Clustering, SMOTE

Backend & Systems: FastAPI (Async Python), Node.js, Express.js, PHP, React 18, Vite, Asynchronous I/O (asyncpg), RESTful APIs, Session Security (Scrypt, OTP)

Databases & DevOps: PostgreSQL, MySQL, Relational Database Architecture, Docker, Linux, Git, GitHub, Postman

TECHNICAL PROJECTS

AI Appointment Assistant & Workflow Automation Engine | *FastAPI, React 18, PostgreSQL, n8n, Google Gemini, Docker* [GitHub Code]

- Architected a production multi-tenant appointment platform combining a React 18 chat portal with an asynchronous FastAPI backend and an autonomous n8n booking workflow powered by Google Gemini.
- Engineered function-calling tools for real-time slot checking and conflict-free database commits; implemented dynamic RFC-5545 `.ics` calendar invite generation and dual-engine SQL fallback resilience.

Autonomous AI Meeting Scheduling Agent | *Python, n8n, Zoom API, Google Calendar API, Gmail API, Docker* [GitHub Code]

- Engineered an autonomous meeting orchestration pipeline using n8n to ingest webhook payloads, negotiate time slots, and provision video conferences.
- Integrated Zoom Server-to-Server OAuth, Google Calendar, and Gmail APIs to automate room creation, invite dispatch, and calendar synchronization.

Customer Segmentation via Hierarchical Clustering | *Python, Scikit-learn, SciPy, Agglomerative Clustering, PCA, Pandas* [GitHub Code]

- Processed 541,000+ transactional line items into RFM customer behavioral feature vectors across 4,372 accounts, applying feature scaling and 2-component PCA (97.39% variance explained).
- Implemented Agglomerative Hierarchical Clustering using Ward's minimum-variance linkage and SciPy dendrograms to classify users into 3 distinct behavioral cohorts.

Financial Fraud Detection & Anomaly Classification System | *Python, Scikit-learn, Imbalanced-Learn, SMOTE, Pandas* [GitHub Code]

- Built an end-to-end anomaly detection pipeline on 284,000+ financial transactions with acute class skew (0.17% fraud prevalence).
- Standardized 30 high-dimensional feature vectors, applied SMOTE to balance minority fraud representation without data leakage, and achieved a **90.82% fraud interception recall rate** with 97.73% overall accuracy.

Full-Stack Hostel Management System | *Node.js, Express.js, MySQL, AWS RDS, Modular MVC* [GitHub Code]

- Developed a full-stack student housing operations platform managing rooms, resident allocations, staff administration, and fee tracking.
- Architected a normalized 25-table relational schema deployed on AWS RDS with strict foreign keys, query indexing, and modular RESTful API endpoints.

International Banking Management System | *Modern C++17, OOP Architecture, File* I/O*, Data Persistence* [GitHub Code]

- Engineered a terminal-based banking system in C++17 applying core OOP principles: class inheritance, polymorphism, and state encapsulation.
- Implemented credential authentication, input validation, inter-account balance transfers, and flat-file transaction audit persistence via CSV serialization.

EDUCATION

FAST National University of Computer and Emerging Sciences (FAST-NUCES) — Peshawar, Pakistan
Bachelor of Science in Computer Science (BSCS) — **Aug 2024 – Jun 2028 (Expected)**
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Operating Systems, Database Systems

Government Sadiq Abbas Postgraduate College — Dera Nawab Sahib, Pakistan
Higher Secondary School Certificate (HSSC) — Pre-Engineering (FSc) — **2022 – 2024**

Secondary School Certificate Examination (Matric) — Bahawalpur, Pakistan
Secondary School Certificate (SSC) — Science — **2020 – 2022**

STRATEGIC PROBLEM SOLVING & HONORS

Competitive Chess (Chess.com: @umair-nawaz): 1300+ Rapid Elo Rating with **4,500+ puzzles solved**, demonstrating advanced algorithmic pattern recognition, forward calculation, and analytical decision-making under time constraints.